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CSA1717 – ARTIFICIAL INTELLIGENCE

CO1 – LIST LAB PROGRAMS PYTHON

EXP 10: A* Algorithm

PROGRAM:

practical lab 10.py - C:/Users/yashm/OneDrive/Desktop/PYTHON AI/practical lab 10.py (3.14.5)

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```
import heapq

graph = {
    "A": {"B": 1, "C": 4},
    "B": {"C": 2, "D": 5},
    "C": {"D": 1},
    "D": {}
}

heuristic = {
    "A": 4,
    "B": 4,
    "C": 1,
    "D": 0
}

def astar(start, goal):
    queue = [(heuristic[start], 0, start, [start])]
    best_cost = {start: 0}

    while queue:
        f, g, node, path = heapq.heappop(queue)

        if node == goal:
            return path, g

        if g != best_cost[node]:
            continue

        for neighbor, cost in graph[node].items():
            new_cost = g + cost

            if new_cost < best_cost.get(neighbor, float("inf")):
                best_cost[neighbor] = new_cost
                new_f = new_cost + heuristic[neighbor]

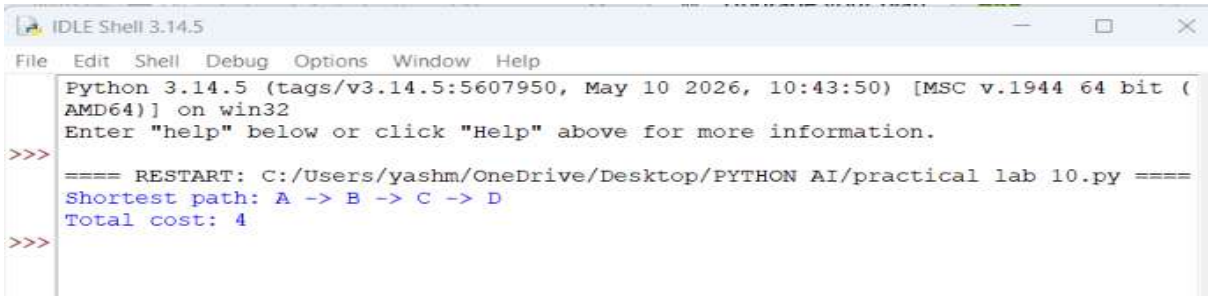
                heapq.heappush(
                    queue,
                    (new_f, new_cost, neighbor,
                     path + [neighbor])
                )

    return None, float("inf")

path, cost = astar("A", "D")

print("Shortest path:", " -> ".join(path))
print("Total cost:", cost)
```

OUTPUT:


IDLE Shell 3.14.5
File Edit Shell Debug Options Window Help
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
==== RESTART: C:/Users/yashm/OneDrive/Desktop/PYTHON AI/practical lab 10.py ====
Shortest path: A -> B -> C -> D
Total cost: 4
>>>